

Retail Customer Behavior & RFM Analysis

Project Objective

The goal of this project is to analyze retail transaction data to understand customer purchasing behavior and identify high-value customers using RFM (Recency, Frequency, Monetary) analysis.

The insights are visualized using a Power BI dashboard.

Business Problem

Retail companies generate large volumes of transaction data. However identifying valuable customers and understanding their purchasing behavior is difficult.

This project converts raw transaction data into actionable insights that help businesses improve customer targeting and retention strategies.

Dataset Overview

The dataset contains online retail transaction records.

Attributes:

Invoice – Transaction ID

StockCode – Product ID

Description – Product name

Quantity – Number of items purchased

InvoiceDate – Purchase date

Price – Price per item

Customer ID – Unique customer identifier

Country – Customer location

```
import pandas as pd  
df = pd.read_excel("online_retail_II_reduced.xlsx")  
df.head()
```

Star Schema Design

The dataset follows a star schema design commonly used in data warehouse systems.

Fact Table

Fact_Sales

- Invoice
- StockCode
- Customer ID
- Quantity
- Price
- InvoiceDate

Dimension Tables

Dim_Customer

- Customer ID
- Country

Dim_Product

- StockCode
- Description

Dim_Date

- InvoiceDate

```
Dim_Customer
|
```

Dim_Product --- Fact_Sales --- Dim_Date

Data Cleaning

The raw dataset contained missing values and invalid transactions.

Cleaning steps performed:

- Removed missing Customer IDs
- Removed negative quantities representing returns
- Converted date columns to datetime format

```
df = df.dropna(subset=['Customer ID'])
df = df[df['Quantity'] > 0]
```

Feature Engineering

A new feature called TotalAmount was created to represent the revenue generated per transaction.

```
df['TotalAmount'] = df['Quantity'] * df['Price']
```

Customer Level Dataset

Transaction data was aggregated at the customer level to prepare data for behavioral analysis.

Each row represents one customer with metrics:

- Recency
- Frequency
- Monetary

RFM Analysis

RFM is a customer segmentation method based on:

Recency – How recently the customer purchased

Frequency – How often the customer purchases

Monetary – How much the customer spends

```
rfm = df.groupby('Customer ID').agg({  
    'InvoiceDate': lambda x: (analysis_date - x.max()).days,  
    'Invoice': 'nunique',  
    'TotalAmount': 'sum'  
})  
  
rfm.columns = ['Recency', 'Frequency', 'Monetary']
```

Customer Segmentation

Customers were segmented into behavioral groups:

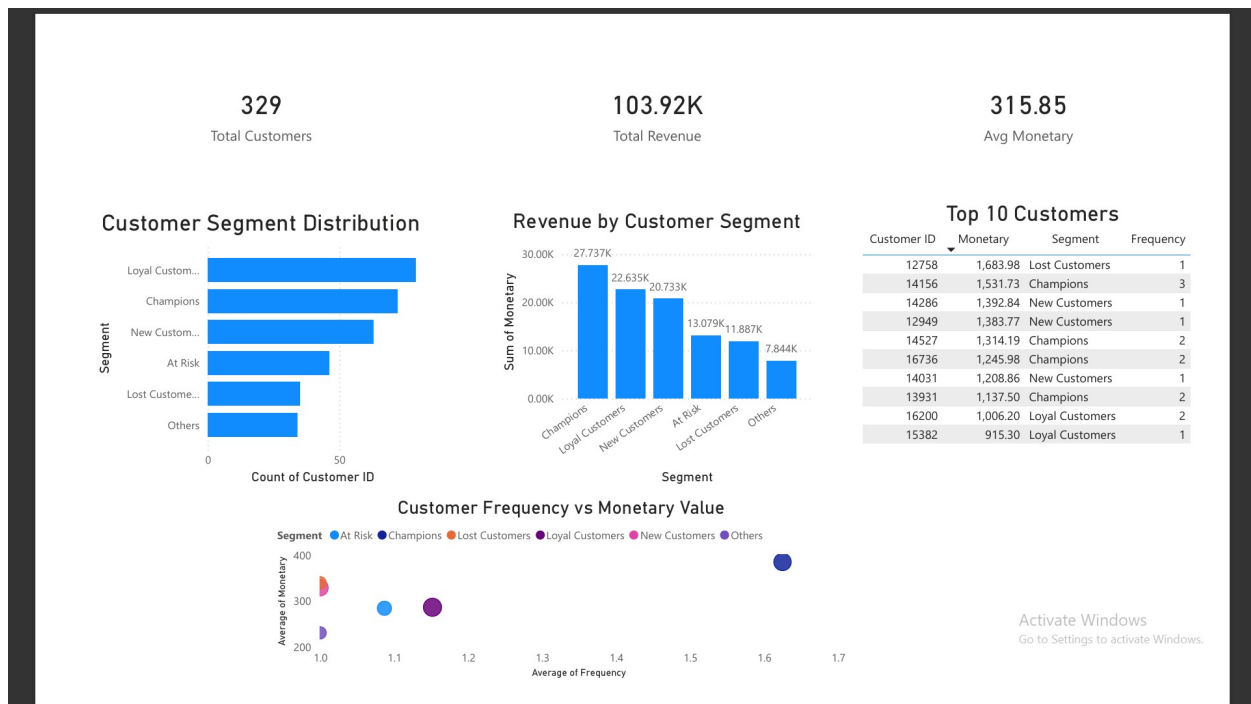
- Champions
- Loyal Customers
- New Customers
- At Risk
- Lost Customers
- Others

Dashboard Visualization

The final insights were visualized using a Power BI dashboard.

The dashboard includes:

- Customer Segment Distribution
- Revenue by Segment
- Top Customers
- Frequency vs Monetary analysis



Conclusion

This project demonstrates how retail transaction data can be transformed into actionable business insights using RFM segmentation and data visualization.